

TrashCam: Real-Time Littering Detection

F24-093-D-TrashCam

Project Team

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Chapter 1

Introduction

This document presents the requirements for TrashCam, a final-year university project focused on creating a real-time litter detection system. TrashCam leverages advanced technologies such as computer vision, machine learning, and IoT to automatically detect and report littering incidents. By analyzing video feeds, identifying faces, and reading license plates, the system provides actionable information to authorities, enabling them to address littering in public spaces effectively.

The document is intended for the following stakeholders:

- **Developers:** Responsible for building and maintaining the system based on the outlined technical and functional requirements.
- **Project Supervisors and Advisors:** To monitor project progress, ensure alignment with academic goals, and provide feedback on the system's development.
- **University Evaluation Panel:** To assess the project based on its technical complexity, innovation, and adherence to requirements.
- **End-Users:** Including citizens reporting litter through the mobile app and authorities using the system to manage incidents and issue fines.
- **Testers:** Tasked with validating that the system functions as expected and meets performance criteria.
- **Documentation Writers:** To produce user manuals and documentation for system deployment and usage.

This document serves as a guide for all stakeholders to ensure the TrashCam project is developed in accordance with the outlined requirements, fulfilling both academic and practical goals.

1.1 Problem Statement

Littering is a pervasive issue in Pakistan, posing serious threats to urban aesthetics, public health, and tourism. Pakistan generates an estimated 36 million tons of solid waste, which is projected to increase to 85 million tons by 2050 due to rapid population growth, urbanization, and economic development[2]. While about 50% of waste is collected, much of the rest is left uncollected, particularly in rural areas, contributing to air, soil, and water pollution[1]. The accumulation of waste in public spaces not only tarnishes the beauty of cities but also leads to environmental degradation, affecting air and water quality. Despite various government initiatives, such as cleanliness drives and awareness campaigns, there is no effective system in place to monitor and control littering on a large scale. TrashCam is designed to fill this gap by offering a fully automated solution to detect, identify, and report instances of littering in real time. Our system leverages advanced technologies like AI for object detection, facial and license plate recognition for identifying perpetrators, and GPS for accurate location tracking of littering events. By integrating with city databases, TrashCam ensures that authorities receive immediate notifications, enabling quick and efficient action. This will not only help in penalizing offenders but also promote a sense of accountability among citizens. Furthermore, the inclusion of a mobile application empowers individuals to report littering incidents, fostering a community-driven approach to maintaining clean public spaces. In the long run, TrashCam aims to create a sustainable environment by reducing waste and encouraging responsible behavior.

1.2 Scope

TrashCam aims to develop a fully automated system that detects littering incidents in public spaces using advanced AI technologies. The system's main functionalities include real-time trash detection through object recognition, offender identification via facial recognition, and vehicle license plate identification to pinpoint individuals involved in littering. The detection system will differentiate between intentional littering and accidental disposal by analyzing behavior, such as whether an item is thrown in a bin or dropped and picked up. If facial recognition fails, feature analysis and cross-referencing family members through number plate detection will aid in identifying the offender. The collected data will be forwarded to a language model (LLM) for report generation, summarizing the incident, including offender details and behavior analysis. The report will then be automatically logged in the crime database, and appropriate fines will be issued. The system targets public areas like parks, streets, and parking lots, with optional integration for private properties or corporate environments. Citizens will also be able to report incidents using a mobile application, fostering community participation. Future developments may expand the system to track illegal dumping, analyze waste management

patterns, and extend its scope to other urban cleanliness challenges.

1.3 Modules

The proposed project consists of several modules divided into two system types: Web App and Mobile App. Each module includes the respective features and functionalities for the web and mobile platforms.

1.3.1 Client Web App

This app is designed for authorities to monitor and take action on littering incidents detected through the system. It consists of the following modules:

1.3.1.1 Litter Detection

This module is responsible for detecting litter in real-time from video or image feeds using advanced object detection techniques.

1. Detect trash objects such as bottles, cans, and papers using YOLO.
2. Analyze behavior to differentiate between intentional littering and accidental drops.
3. Visualize detected litter on a live map with GPS data.
4. Generate real-time alerts for immediate action.

1.3.1.2 Number Plate Reading

This module is responsible for reading vehicle license plates using Optical Character Recognition (OCR) technology.

1. Capture and extract vehicle license plate numbers from images or video footage.
2. Cross-reference the license plate number with government databases to identify the vehicle owner.
3. Log vehicle data for record-keeping and further investigation.

1.3.1.3 Facial Recognition

This module utilizes advanced facial recognition technology to identify individuals responsible for littering.

1. Capture and analyze facial features from video or image feeds.
2. Match the detected faces with known databases for offender identification.
3. Alert authorities if an individual is identified.

1.3.1.4 Report Generation and Fine Issuing

This module automatically generates reports and issues fines after detecting littering incidents.

1. Generate detailed reports including the type of litter, location, and time of the incident.
2. Automatically issue fines based on the severity of the offense using the collected data.
3. Send notifications to offenders regarding issued fines.

1.3.2 Client Mobile App

This app is designed for citizens to report littering incidents and contribute to a cleaner environment. It consists of the following modules:

1.3.2.1 Litter Reporting

This module allows users to report littering incidents through the mobile app.

1. Capture and upload images or videos of littering incidents.
2. Provide details like location, type of litter, and time of the incident.
3. Submit reports to the authorities for action.

1.3.2.2 License Plate Reporting

This module allows users to report vehicles involved in littering incidents.

1. Capture vehicle license plates involved in littering.
2. Submit the report with the license plate details, including vehicle make and model if available.
3. Notify authorities for further investigation.

1.3.2.3 Incident History

This module enables users to track the status of their reports and receive updates.

1. View the history of submitted littering reports.
2. Check the status of ongoing investigations and actions taken.
3. Receive updates on whether fines were issued or actions were taken.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User class	Description
Municipal Authorities	These are local government bodies responsible for public sanitation and waste management. They will use the system to receive real-time reports on littering incidents, monitor public spaces such as parks and streets, and issue fines or warnings based on offender identification via facial or license plate recognition. They are key users who will make decisions on deploying resources to clean up littering hotspots.
Law Enforcement Agencies	Police departments and traffic control units that will enforce littering laws. They will use the system's facial recognition and vehicle number plate identification features to trace offenders involved in littering. Reports generated by the system will help them take appropriate legal actions.
Citizens	General public who will use the mobile app to report littering incidents. They will be able to submit photos, videos, and locations of littering events, contributing to the community's cleanliness. They will also be able to track incidents and see actions taken by authorities.
Waste Management Companies	Companies involved in waste collection and disposal. They will use the system's reports to optimize collection routes, focus on high-littering areas, and improve operational efficiency. They will benefit from data-driven insights to deploy resources more effectively.
Environmental Agencies	NGOs and environmental protection organizations that will use the system to promote environmental responsibility. They will access data insights to identify recurring littering issues and create targeted campaigns for awareness and cleanliness initiatives.

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter outlines the functional and non-functional requirements for the TrashCam project, which are critical to its design and development. The functional requirements specify the system's key features, such as litter detection, offender identification, report generation, and fine issuing, broken down into various modules.

Non-functional requirements cover essential system qualities like performance, reliability, security, and usability, ensuring the system's overall efficiency and scalability.

Additionally, this chapter introduces modeling techniques, such as use case diagrams, event-response tables, and domain models, to help visualize the system's structure and operations. These models are essential for both developers and stakeholders to understand how different components of TrashCam interact and work together.

By addressing these requirements and models, this chapter provides a clear roadmap for the successful development of TrashCam, ensuring it meets both user and technical expectations.

2.1 Use-case/Event Response Table/Storyboarding

For TrashCam, the requirement gathering involves a combination of use cases and event-response tables:

2.1.1 Use Case Diagrams

TrashCam has interactive end-user components, particularly through the mobile application where users report littering incidents. Thus, a use-case diagram can be created for mobile user interaction.

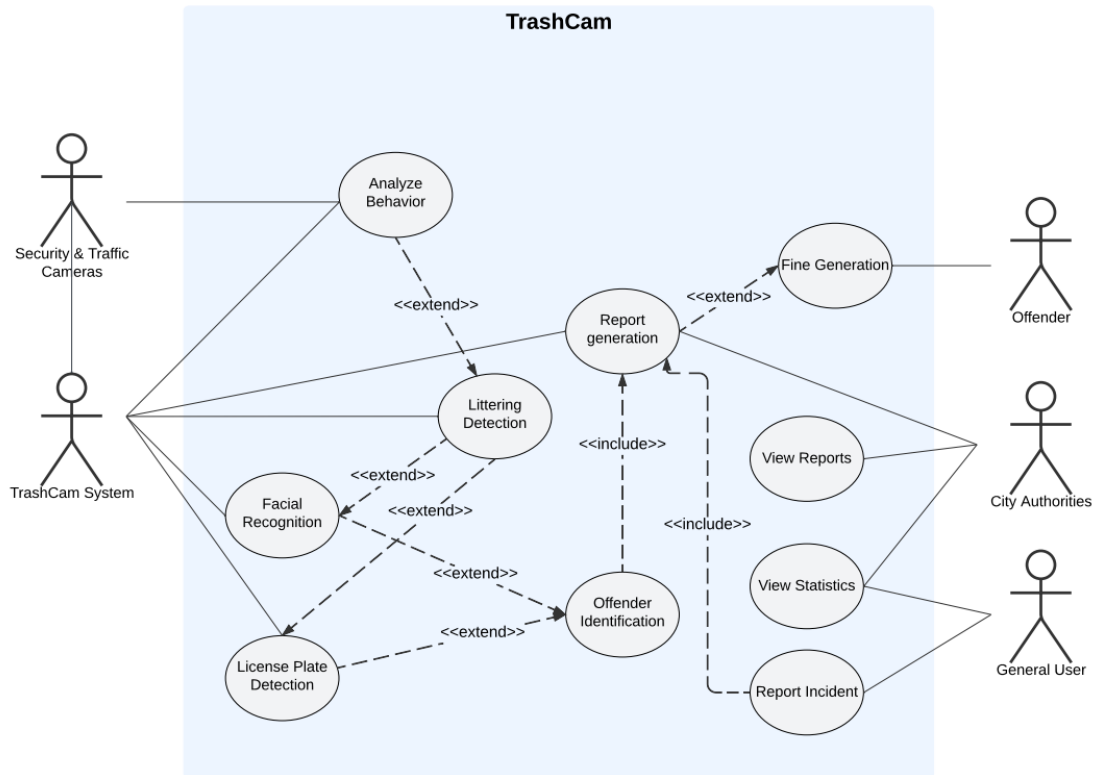


Figure 2.1: Use Case Diagram for TrashCam Mobile User Interaction

2.1.2 Event-Response Table

This technique will be useful for the real-time litter detection system which involves multiple backend processes like detecting litter, identifying offenders, and issuing reports.

Event	Response
Litter detected by camera	System captures video footage and processes it through YOLO to identify trash objects (e.g., bottles, cans).
Object classified as litter	System flags the object and initiates behavior analysis to determine if littering was intentional or accidental.
Behavior identified as intentional	System captures the offender's facial features using facial recognition.
Offender's face detected	System compares the detected face with existing databases of known offenders.
Offender's face not detected	System proceeds to identify vehicle license plate using OCR (if the offender was near or in a vehicle).
License plate identified	System cross-references the license plate with a government database to identify the owner of the vehicle.
Offender identified	System generates a report including offender details, litter type, location, and timestamp.
Report generated	The report is logged in the city's crime database and authorities are notified.
Fine determined based on offense	A fine is automatically issued and sent to the offender.
Citizen reports litter via mobile app	System logs the report, verifies the location and time, and generates a notification for authorities.

2.2 Functional Requirements

The functional requirements for TrashCam are categorized into different modules.

2.2.1 Module 1: Litter Detection

1. FR1: The system shall detect litter objects such as bottles, cans, and papers using the YOLO model.
2. FR2: The system shall detect the location of the incident using the Google Maps API.

3. FR3: The system shall classify the litter into predefined categories (e.g., plastic, glass, metal, organic waste) to adjust fines based on the state's legal regulations and the environmental impact of the litter.

2.2.2 Module 2: Behavior Analysis

1. FR1: The system shall differentiate between intentional littering and accidental disposal based on behavior analysis.
2. FR2: The system shall detect vehicle-related littering by analyzing the motion of individuals throwing objects out of vehicles, correlating the action with vehicle movement, and assessing if the vehicle is stationary or moving.

2.2.3 Module 3: Number Plate Recognition

1. FR1: The system shall recognize and read vehicle license plates using Optical Character Recognition (OCR).
2. FR2: The system shall cross-reference the license plates with government databases to identify vehicle owners.

2.2.4 Module 4: Facial Recognition

1. FR1: The system shall capture and analyze facial features from video footage.
2. FR2: The system shall match the captured faces with known offender databases for identification.
3. FR3: The system shall trace the close relatives of the offender using available public databases or government records to confirm the identity.

2.2.5 Module 5: Report Generation and Fine Issuing

1. FR1: The system shall generate detailed reports for each littering incident, including the type of litter, location, time, identity of the offender, and how the event took place.
2. FR2: The system shall include any photographic or video evidence in the report, including frames from the footage showing the littering action and offender identification.

3. FR3: The system shall cross-reference the incident location with local zoning laws or environmental protection regulations, highlighting any additional penalties for littering in restricted or sensitive areas.
4. FR4: The system shall log and archive all reports in a centralized database for future reference, with search filters for location, offender, litter type, or time period.
5. FR5: The system shall notify responsible authorities in real-time, sending reports via email, SMS, or through a designated portal for immediate response depending on the urgency of the incident.

2.2.6 Module 6: Fine Issuing

1. FR1: The system shall issue fines based on the offense's severity to the offender.
2. FR2: The system shall notify the offender of their fine through multiple channels (e.g., SMS, email, mail) including details of the offense, supporting evidence (photos or video), and instructions for payment.
3. FR3: The system shall include an analytics dashboard for authorities to view statistics on fine issuance, payment rates, and geographical trends in littering offenses, helping inform public policies and initiatives.

2.3 Non-Functional Requirements

2.3.1 Reliability

1. R1: The system should ensure a mean time between failures (MTBF) of at least 200 hours, ensuring reliable detection, analysis, and reporting during live operations.
2. R2: The system should store incident data in a backup database to avoid data loss in case of system failure.
3. R3: The system should be capable of handling up to 50 simultaneous detections without interruption or failure.

2.3.2 Usability

1. USE-1: The system's web and mobile interfaces should be intuitive and user-friendly, allowing users to generate reports and view data with minimal guidance.

2. USE-2: The mobile app should allow users to report littering incidents with one click, optimizing simplicity for end-users.
3. USE-3: The system should be designed with basic accessibility features such as clear visual elements to support easy navigation for a diverse user base.

2.3.3 Performance

1. PER-1: The system should detect and classify littering events within 5 seconds of occurrence to ensure near real-time response.
2. PER-2: The system should generate reports within 10 seconds after detecting a littering event.
3. PER-3: The system should support up to 10 concurrent users accessing data without noticeable performance degradation.

2.3.4 Security

1. SEC-1: The system should implement basic authentication to protect sensitive data such as incident reports and offender identity.
2. SEC-2: Data should be transmitted using secure HTTPS protocols to ensure confidentiality during communication.
3. SEC-3: Personal data (e.g., offender details) should be anonymized to comply with privacy standards.
4. SEC-4: The system should provide role-based access ensuring that only authorized users can access specific features (e.g., report management or fine issuance).

2.4 Domain Model

The domain model represents the key entities, their attributes, and relationships within the TrashCam system. This model helps visualize how different components of the system, such as users, cameras, offenders, and reports, interact with each other and how data is stored and processed.

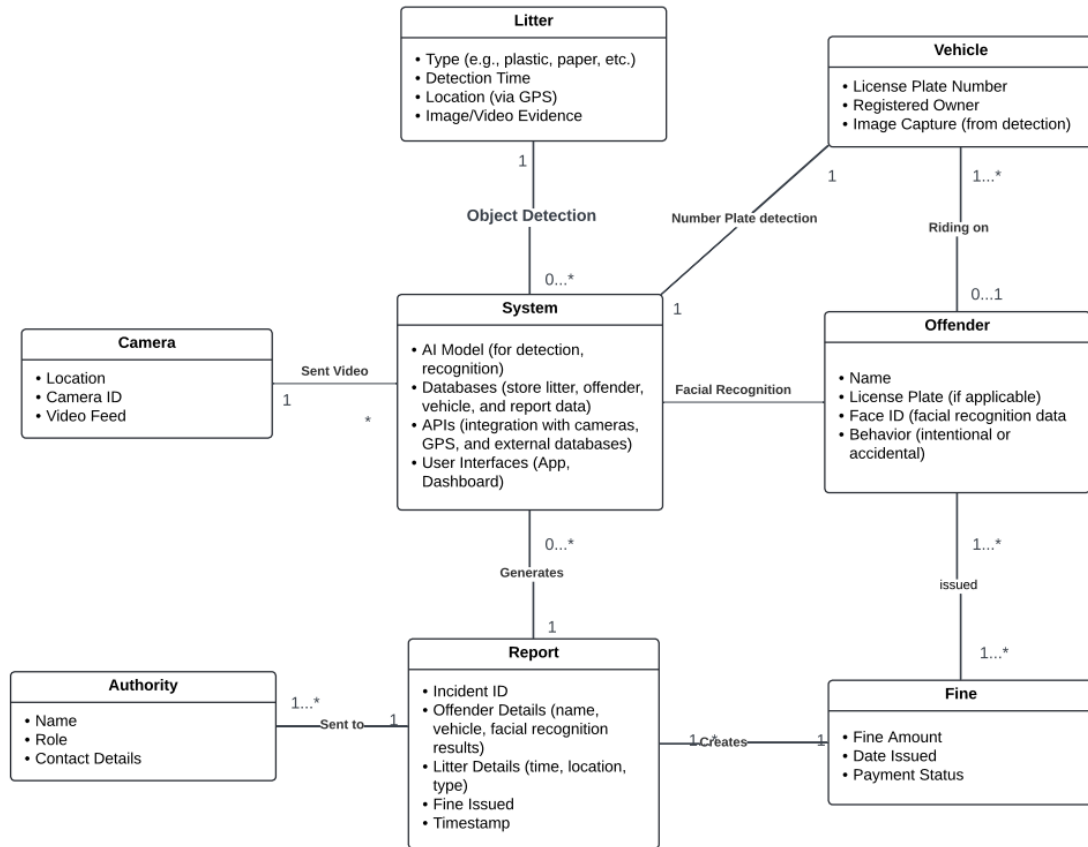


Figure 2.2: Domain Model for TrashCam System

Chapter 3

System Overview

3.1 Architectural Design

The TrashCam system is designed using the Model-View-Controller architecture pattern to ensure a clear separation of concerns for processes such as litter detection, offender identification, and report generation. The architecture consists of three primary components:

- **Model:** Manages the data, business logic, and database interactions. This includes the storage and management of information about users, cameras, reports, offenders, and authorities.
- **View:** Represents the user interface elements, provided through both the mobile and web applications. Users interact with the system via the View, reporting littering incidents, viewing data, and receiving notifications.
- **Controller:** Acts as the intermediary between the Model and the View. It handles user input, processes data from cameras and other sources, and updates the View with information from the Model.

The MVC architecture ensures that each component operates independently, promoting scalability, maintainability, and ease of testing. Modules such as image processing, facial recognition, and license plate detection interact through the Controller layer with the backend Models, ensuring a well-organized and efficient system.

First, the Box and Line Diagram provides a simple representation of the system's main components and their relationships:

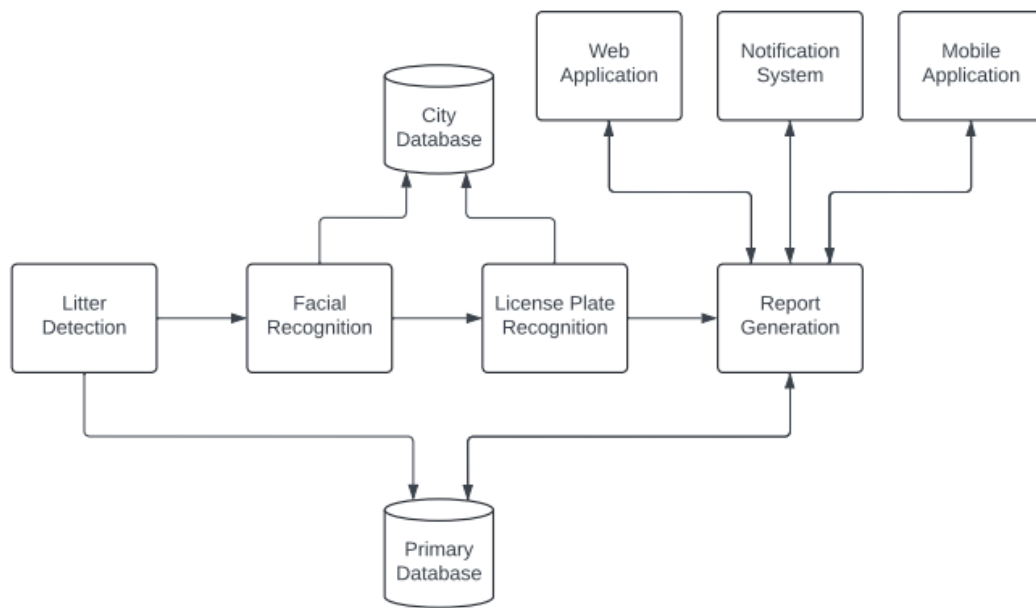


Figure 3.1: Box and Line Diagram for TrashCam System

Next, the following diagram illustrates the final architecture of the TrashCam system:

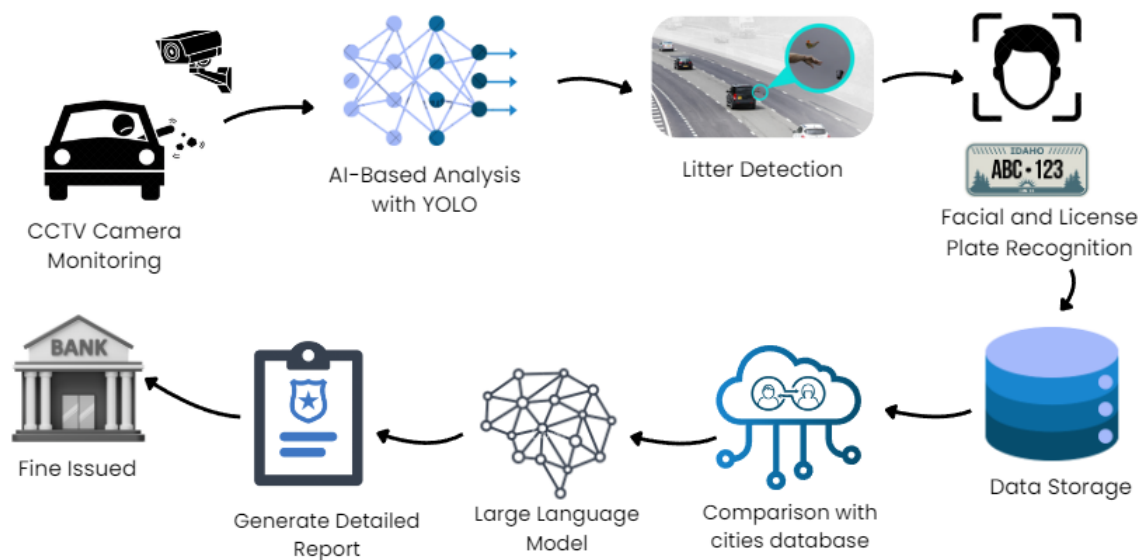


Figure 3.2: TrashCam System Architecture

3.2 Design Models

3.2.1 Activity Diagram

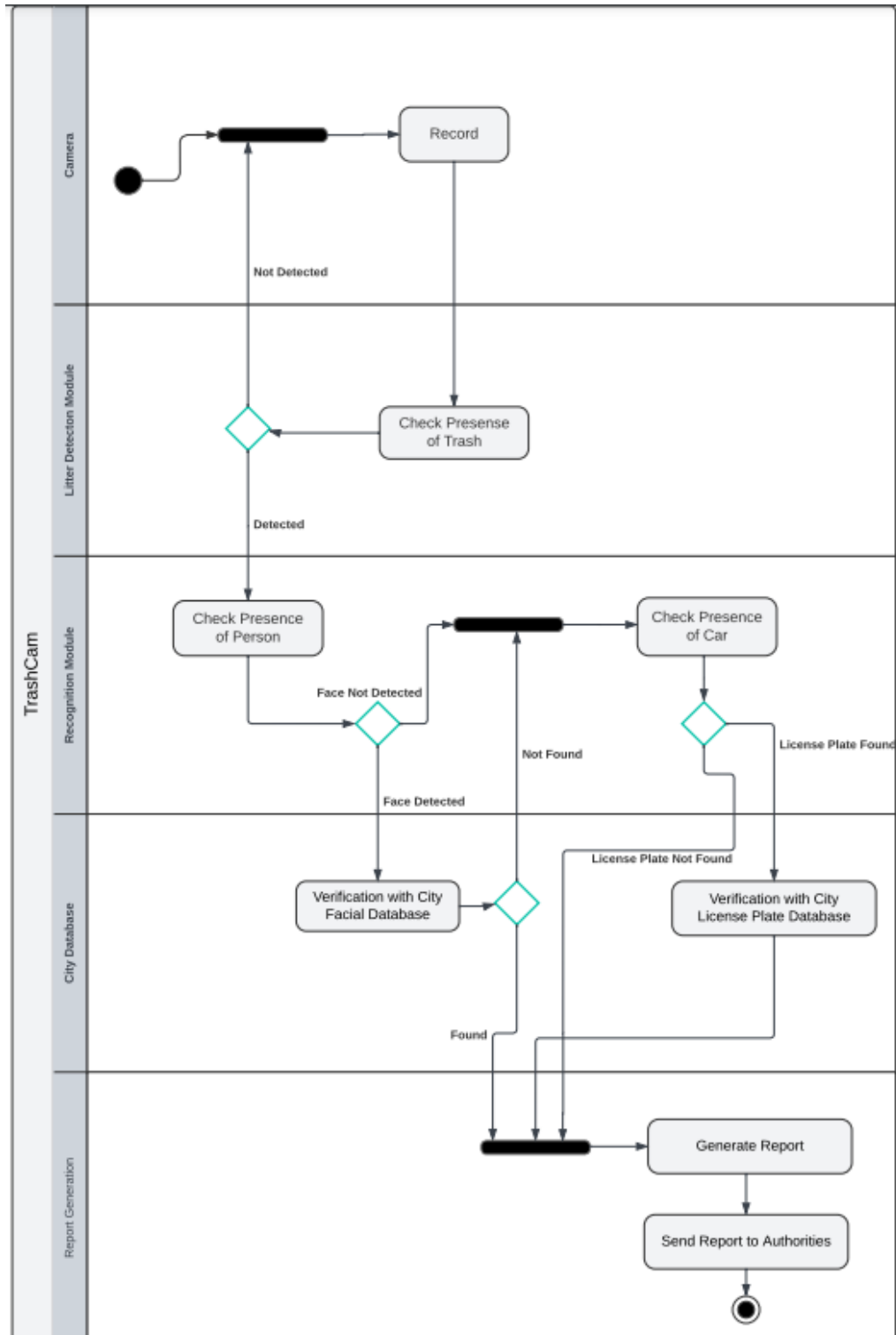


Figure 3.3: Activity Diagram for Litter Detection and Reporting Process

3.2.2 Class Diagram

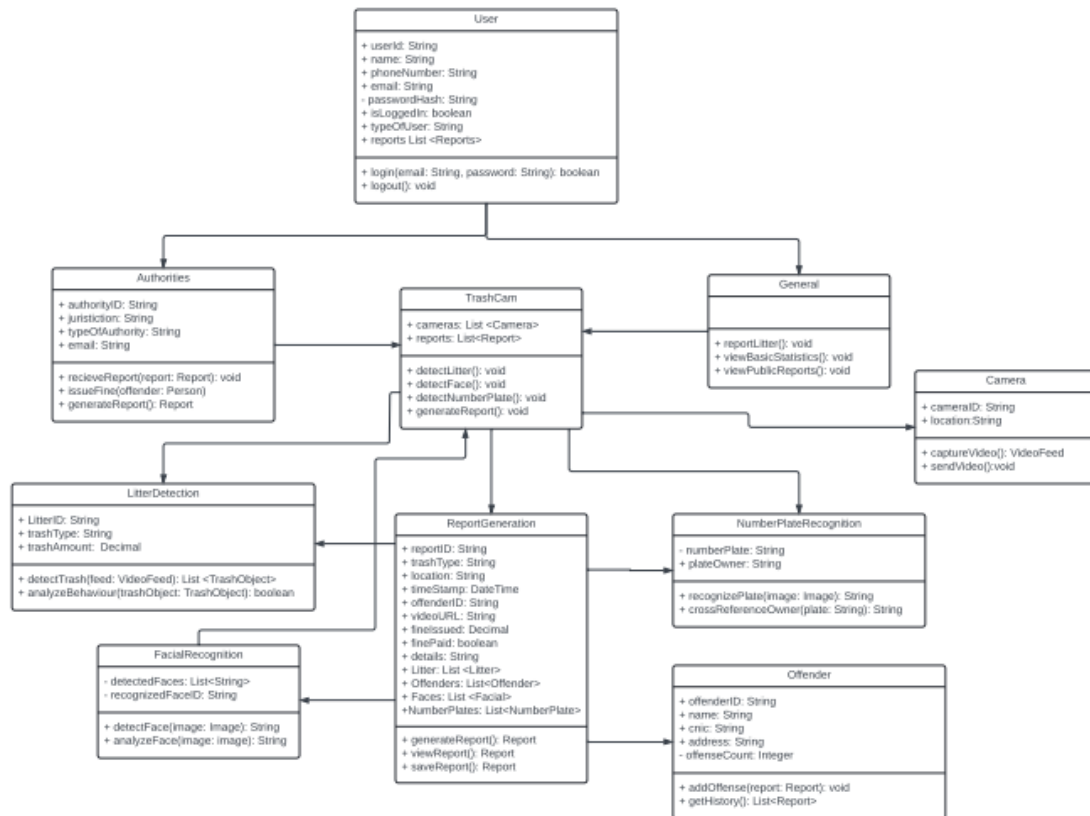


Figure 3.4: Class Diagram of the System

3.2.3 State Transition Diagram

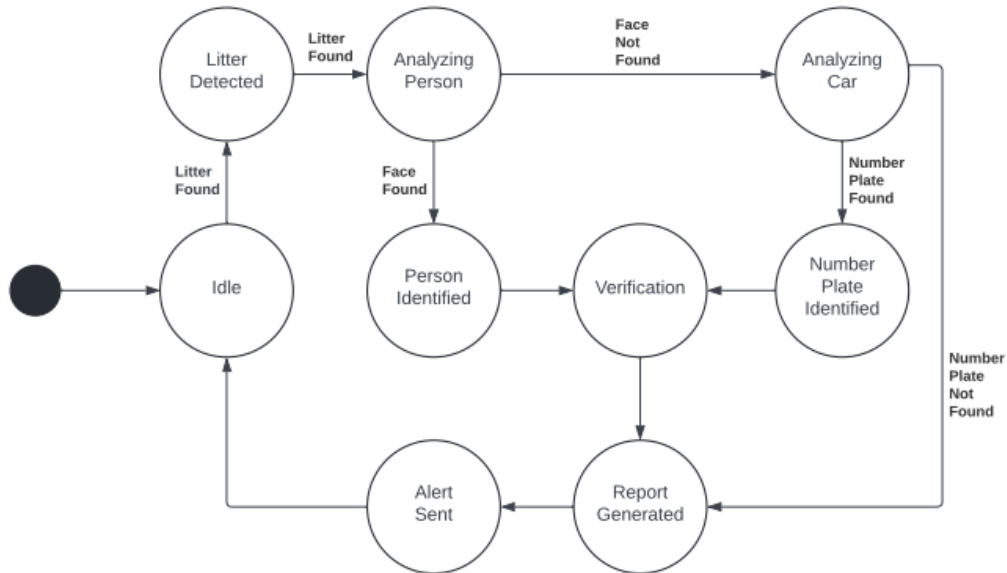


Figure 3.5: State Transition Diagram for Litter Detection Process

3.2.4 Sequence Diagram

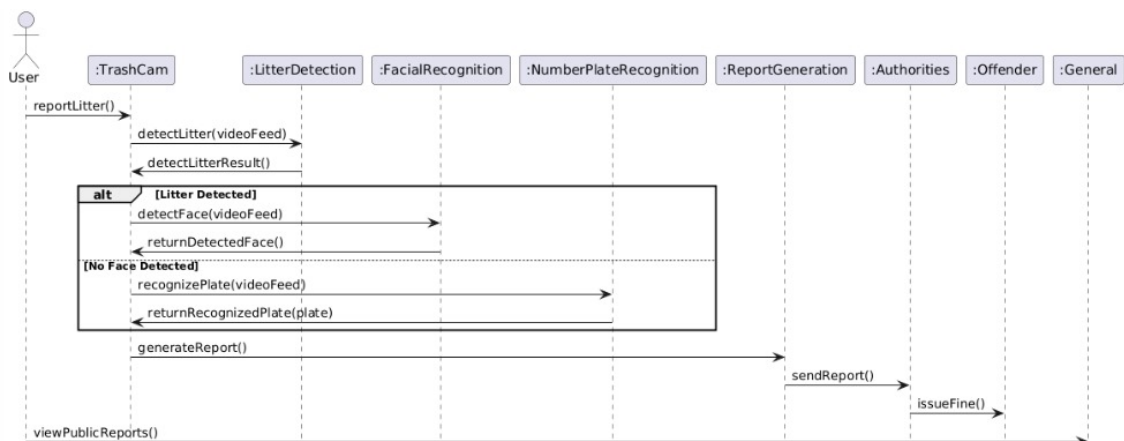


Figure 3.6: Sequence Diagram for TrashCam Litter Detection Process (Part 1)

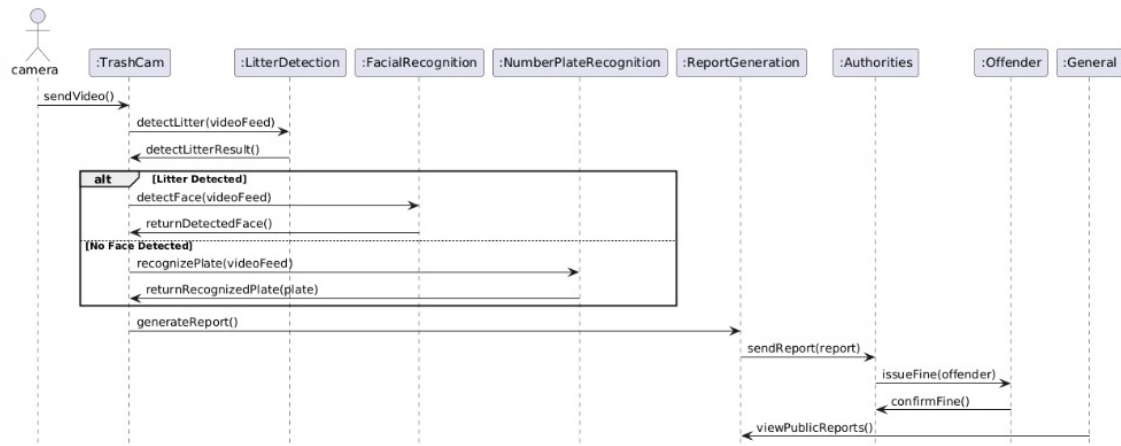


Figure 3.7: Sequence Diagram for TrashCam Litter Detection Process (Part 2)

3.3 Data Design

This section explains how the information domain of the TrashCam system is transformed into data structures. The major entities such as users, cameras, authorities, offenders, reports, and litter events are stored, processed, and organized in a structured manner. The database is designed to store information efficiently, ensuring quick retrieval and data integrity.

3.3.1 Entity-Relationship Diagram

The following diagram shows the Entity-Relationship (ER) model of the TrashCam system, highlighting the main entities and their relationships.

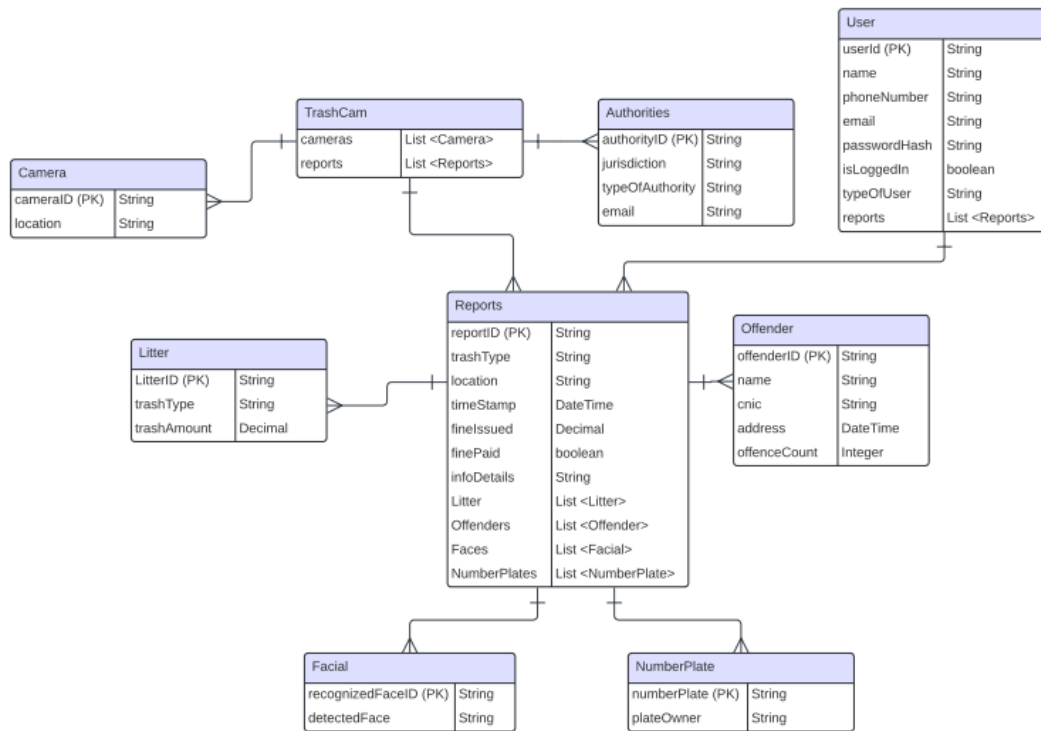


Figure 3.8: Entity-Relationship Diagram of the TrashCam Database

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- [2] World Bank Group. Pakistan: Solid waste management for a circular economy - a roadmap. Technical report, World Bank, 2020.